

PS5: Does a Covered Call Fund Beat Its Shares?

CHEME 4/5660 Cornell University | Mathematical Companion

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Training only. This assignment uses historical option quotes, a simple price model, and simplified trading assumptions for teaching. It provides no investment advice.

Covered calls. A covered call fund keeps the full loss of its shares, less the premiums, and gives up every gain above each month's strike. A collar limits each month's loss with a put, at the cost of that put's premium.

Simplifying assumptions. One call (and, for the collar, one put) per share, measured per share. Every option is held to expiration and settled at the close that day, with no early exercise. We omit commissions, dividends, and taxes. The quotes come from an indicative data feed.

The central question

Does a covered call fund beat simply holding the shares, and what does it give up?

- Study 1: the promise. Simulate six months of share prices for each ticker, price the calls on the binomial tree, and compare the distributions of the fund's and the shares' growth rates.
- Study 2: what happened. Replay each ticker's six real monthly calls at their market prices, and compare the fund with the shares, SPY, and cash over the same days.
- Study 3: the conclusion. Decide what the fund gave up and what it got, and what one half-year of history can tell us.

As in the course example, S_j is the share price at the start of month j, K_j the strike of the call sold then, and $\mathcal{P}_{c,j}$ its premium (USD/share).

Reading guide Both tracks do Studies 1--3. The collar fund on slide 11 is Advanced only.

Pricing the monthly call

The call sold at the start of month j, strike K_j, expires after Δt years. We price it on the Cox–Ross–Rubinstein (CRR) binomial tree with $h = 60$ steps of length $\tau = \Delta t / h$:

$$u = e^{\sigma\sqrt{\tau}}, \qquad d = \frac{1}{u}, \qquad \bar{p} = \frac{e^{g_y \tau} - d}{u - d},$$

where σ is the pricing volatility and g_y the continuously compounded risk-free rate. Rolling the payoff $(S - K_j)^+$ back through the tree under $\bar{p}$, discounting by $e^{-g_y \tau}$ per step and keeping at each node the larger of that value and the exercise value (American), gives $\mathcal{P}_{c,j}$.

In code: `build(MyAmericanCallContractModel, ...)` with strike K_j, then `binomial_premium(call; method = :crr, h = 60, ...)` with time Δt, volatility σ, share price S_j, and rate g_y.

No dividends An American call on a stock without dividends is never worth exercising early, so it is priced like a European call.

One month of the covered call fund

The fund's wealth W_j is its shares at S_j. It sells one call per share and holds the premium as cash, which grows by $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$ over the call's life T. At expiration the shares are worth S_{j+1}, less the call's payoff:

$$\frac{W_{j+1}}{W_j} = \frac{S_{j+1} - \left(S_{j+1} - K_j\right)^+ + \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_{c,j}}{S_j} = \frac{\min\left(S_{j+1}, K_j\right) + \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_{c,j}}{S_j}.$$

Holding the shares instead gives S_{j+1}/S_j.

- Expires worthless, $S_{j+1} \le K_j$: the fund leads the shares by $\mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_{c,j}/S_j$.
- Exercised, $S_{j+1} > K_j$: the ratio stops at $(K_j + \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_{c,j})/S_j$.

Test month $S_j = 100$, $K_j = 102$, $\mathcal{P}_{c,j} = 0.87$, $\mathcal{D} = 1.00345$: 1.0053 at $S_{j+1} = 99.66$; 1.0287 for any $S_{j+1} \ge 102$.

The growth rate over a roll

Over N months that last T_{roll} years in all, the fund's growth rate is the log of its final over its initial wealth, divided by the time. The final wealth is the product of the monthly ratios, so:

$$g = \frac{1}{T_{\text{roll}}} \ln\left(\frac{W_{N+1}}{W_1}\right) = \frac{1}{T_{\text{roll}}} \sum_{j=1}^{N} \ln\left(\frac{W_{j+1}}{W_j}\right) \qquad \text{(units: } 1/\text{yr)}.$$

The same formula with S_{j+1}/S_j gives the shares' growth rate. The order of the ratios does not matter, and a ratio of exactly 1 adds nothing.

- In the model: $N = 6$ months of $\Delta t = D_1/365$ year, the first trade's term, so $T_{\text{roll}} = 6\,\Delta t$.
- In history: T_{roll} is the calendar days from a ticker's first entry to its last expiration, divided by 365, about 0.47 year.

The model of the share price

For each ticker, the model simulates the share price at the start of each month with geometric Brownian motion (GBM), our belief about the share price:

$$S_{j+1} = S_j \exp\left[(\hat{\mu} - \tfrac{\hat{\sigma}^2}{2})\Delta t + \hat{\sigma}\sqrt{\Delta t}\, Z_j\right], \qquad Z_j \sim \mathcal{N}(0,1),$$

on 10,000 paths from $S_1 = S(0)$, the close on the first entry date, with $\Delta t = D_1/365$, the first trade's term. The drift $\hat{\mu} = \hat{g} + \hat{\sigma}^2/2$ and volatility $\hat{\sigma}$ come from the 252 daily changes before that date ($\hat{g}$: mean log growth rate).

The strike is $K_j = m\, S_j$, with $m = K/S(0)$ from the first trade. The call is priced at one of two volatilities:

- historical, $\sigma = \hat{\sigma}$, as in the course example;
- implied, $\sigma = \sigma_{\text{IV}}$, implied by the call's market price on the first entry date.

Two probabilities Paths use our belief (drift $\hat{\mu}$); premiums use the risk-neutral tree (drift g_y).

The premium is a fixed fraction of the share price

Every node of the tree is S_j times a factor that does not depend on S_j, and the payoff $(S - mS_j)^+$ scales the same way. So the premium is proportional to the share price:

$$\mathcal{P}_{c,j} = c(m,\sigma)\, S_j, \qquad c(m,\sigma) = \frac{\mathcal{P}_c\big(S(0),\, m\, S(0)\big)}{S(0)}.$$

The report prices one call per ticker and pricing volatility, and the monthly ratio depends only on that month's price change:

$$\frac{W_{j+1}}{W_j} = \min\left(\frac{S_{j+1}}{S_j},\, m\right) + \mathcal{D}_{T,0}(g_y)\cdot c(m,\sigma).$$

The fund gives up every monthly price ratio above m, and collects $c(m,\sigma)$ every month. A farther strike (larger m) gives up less and collects less.

Report *Premiums: model and market* lists $c(m,\hat{\sigma})$, $c(m,\sigma_{\text{IV}})$, and the market's mid price over $S(0)$.

Replaying history

Each ticker has six back-to-back trades. Trade j enters at the close $S_{0,j}$, sells the call with strike K_j at its mid price $\mathcal{P}_{c,j}$, and settles at the close $S_{T,j}$ on the expiration date, D_j calendar days later:

$$\frac{W_{j+1}}{W_j} = \frac{\min\left(S_{T,j}, K_j\right) + \mathcal{D}_j \cdot \mathcal{P}_{c,j}}{S_{0,j}}, \qquad \mathcal{D}_j = e^{g_y D_j/365}.$$

When trade $j+1$ enters after trade j expires, the fund holds only its shares in between, so the fund and the shares both gain the ratio $S_{0,j+1}/S_{T,j}$. The risk-free rate comes from the 13-week bill yield $y = 0.0376$ and term $\tau = 91/365$, as in the course example:

$$g_y = \frac{1}{\tau}\ln\left(1 + y\,\tau\right) \approx 0.0374 \text{ per year.}$$

Benchmarks SPY bought at its close on the ticker's first entry date and sold on its last expiration date; cash grows at g_y, so its growth rate is g_y.

Promise versus outcome

Here the simulation follows ticker i's real roll: each trade keeps its real $K_j/S_{0,j}$, $\mathcal{P}_{c,j}/S_{0,j}$, and D_j, with the gaps; only prices are simulated. It gives $N_p = 10{,}000$ pairs $(g^{(p)}_{\text{fund}}, g^{(p)}_{\text{shares}})$, and with the indicator $\mathbf{1}[\cdot]$:

$$\hat{P}_i = \frac{1}{N_p}\sum_{p=1}^{N_p} \mathbf{1}\Big[g^{(p)}_{\text{fund}} > g^{(p)}_{\text{shares}}\Big], \qquad \text{expected count} = \sum_{i=1}^{29} \hat{P}_i.$$

For one ticker, with the real difference $\delta_i = g_{\text{fund}} - g_{\text{shares}}$, the model percentile is:

$$q_i = \frac{1}{N_p}\sum_{p=1}^{N_p} \mathbf{1}\Big[g^{(p)}_{\text{fund}} - g^{(p)}_{\text{shares}} \le \delta_i\Big].$$

Reading A percentile near 0% or 100% is rare under the model; with matched trades it points at the share prices.

Advanced: one month of the collar fund

The collar fund also buys a put with strike $K_1 < K_2$, where K_2 is the call's strike. With the put's premium $\mathcal{P}_1$ and the call's $\mathcal{P}_2$, the month's payoffs add to the share price held between a floor and a cap:

$$S_{j+1} + \left(K_1 - S_{j+1}\right)^+ - \left(S_{j+1} - K_2\right)^+ = \min\Big(\max\left(S_{j+1}, K_1\right),\, K_2\Big).$$

The net premium $\mathcal{P}_2 - \mathcal{P}_1$, negative when the put costs more, grows to expiration with the cash:

$$\frac{W_{j+1}}{W_j} = \frac{\min(\max\left(S_{j+1}, K_1\right), K_2) + \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 - \mathcal{P}_1)}{S_j}.$$

The put is the trade's `long_put` leg, near 0.10 delta, well below the share price. The model uses $K_1 = m_1 S_j$ and $K_2 = m\, S_j$, with m_1 and m the first trade's strikes over $S(0)$, and prices the put at its implied volatility.

Course example *A Protective Collar on SPY* uses the same floor K_1, cap K_2, and premiums $\mathcal{P}_1$ (put) and $\mathcal{P}_2$ (call).

Summary

PS5 runs a covered call fund twice: in a model of the share price, and on the real monthly calls of 29 tickers, and compares it with holding the shares.

Key takeaways

- Each month the fund's wealth ratio is the share price capped at the strike plus the grown premium, over the starting price. The premiums lift the falling months and the strike caps the rising ones.
- The growth rate over a roll is the sum of the logs of the monthly wealth ratios divided by the time in years, in 1/yr for the model and for history.
- A model's probability that the fund beats the shares is a promise about many possible half-years; history gives one half-year, shared by all 29 tickers.

Use these measures to answer the central question.